

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE329	PRELUDE TO COMPETITIVE CODING	2	0	1	3	
Course Weightage	ATT: 5 CAP: 45 ETP: 50					
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT,ENTREPRENEURSHIP					

Course Outcomes :Through this course students should be able to

CO1 :: relate the theoretical as well as practical knowledge to form an amalgamation of working code

CO2 :: employ a working combination of solution to ubiquitous problems which are time and space efficient

CO3 :: identify and comprehend the inner workings behind the design of an optimal solution

CO4 :: illustrate the usage of algorithms and data structures in the design of an optimal solution towards a problem

CO5 :: recall the knowledge obtained from various algorithmic paradigms to formulate optimal solutions to real world problems

CO6 :: evaluate the knowledge obtained from various algorithmic paradigms to formulate optimal solutions to real world problems

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	CRACKING THE CODING INTERVIEW	GAYLE LAAKMANN MCDOWELL	CAREERCUP
R-2	DATA STRUCTURES AND ALGORITHMS : CONCEPTS, TECHNIQUES AND APPLICATIONS	G. A. V. PAI	MCGRAW HILL EDUCATION

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://www.codechef.com/	Programming, Data Structures and Algorithms
RW-2	https://www.hackerrank.com/	Programming, Data Structures and Algorithms
RW-3	https://www.hackerearth.com/	Programming, Data Structures and Algorithms
RW-4	https://algorithms.tutorialhorizon.com/	Programming and Algorithms
RW-5	https://www.geeksforgeeks.org/	Programming, Data Structures and Algorithms

LTP week distribution: (LTP Weeks)	
Weeks before MTE	14
Weeks After MTE	0
Spill Over (Lecture)	4

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Basic Data Structures (Arrays, Declaring and processing 1D and 2D arrays)	R-1 R-2	RW-2 RW-4 RW-5	Arrays, Declaring and processing 1D and 2D arrays, Insertion in an array, Deletion from array, Array rotations, Array arrangement, rearrangement	The students will be able understand in depth about the fundamental data structure namely array and will be able to solve problems based on the same.	Live demonstration on compiler	
		Basic Data Structures (Insertion in an array)	R-1 R-2	RW-2 RW-4 RW-5	Arrays, Declaring and processing 1D and 2D arrays, Insertion in an array, Deletion from array, Array rotations, Array arrangement, rearrangement	The students will be able understand in depth about the fundamental data structure namely array and will be able to solve problems based on the same.	Live demonstration on compiler	
		Basic Data Structures (Deletion from array)	R-1 R-2	RW-2 RW-4 RW-5	Arrays, Declaring and processing 1D and 2D arrays, Insertion in an array, Deletion from array, Array rotations, Array arrangement, rearrangement	The students will be able understand in depth about the fundamental data structure namely array and will be able to solve problems based on the same.	Live demonstration on compiler	
		Basic Data Structures(Array rotations)	R-1 R-2	RW-2 RW-4 RW-5	Arrays, Declaring and processing 1D and 2D arrays, Insertion in an array, Deletion from array, Array rotations, Array arrangement, rearrangement	The students will be able understand in depth about the fundamental data structure namely array and will be able to solve problems based on the same.	Live demonstration on compiler	

Week 1	Lecture 1	Basic Data Structures(Array arrangement, rearrangement)	R-1 R-2	RW-2 RW-4 RW-5	Arrays, Declaring and processing 1D and 2D arrays, Insertion in an array, Deletion from array, Array rotations, Array arrangement, rearrangement	The students will be able understand in depth about the fundamental data structure namely array and will be able to solve problems based on the same.	Live demonstration on compiler	
	Lecture 2	Basic Data Structures (Matrix multiplication, Lower triangular and upper triangular matrix of array, Different operations on Matrices)	R-1 R-2	RW-2 RW-3 RW-4 RW-5	Matrix multiplication, Lower triangular and upper triangular matrix of array, Different operations on Matrices, Print a matrix in spiral form, Find distinct elements common to all rows in a matrix	The students will be able understand in depth about the fundamental data structure namely array and will be able to solve problems based on the same.	Live demonstration on compiler	
		Basic Data Structures(Print a matrix in spiral form, Find distinct elements common to all rows in a matrix)	R-1 R-2	RW-2 RW-3 RW-4 RW-5	Matrix multiplication, Lower triangular and upper triangular matrix of array, Different operations on Matrices, Print a matrix in spiral form, Find distinct elements common to all rows in a matrix	The students will be able understand in depth about the fundamental data structure namely array and will be able to solve problems based on the same.	Live demonstration on compiler	
Week 2	Lecture 3	Basic Data Structures(String declaration and manipulation)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	String declaration and manipulation, K maximum sum from two arrays, Missing characters to make a string Pangram, Rearrange characters so that no two adjacent characters are same, Remove minimum number of characters so that two strings become anagram	The students will be able understand in depth about the fundamentals of strings and will be able to solve problems based on the same.	Live demonstration on compiler	

Week 2	Lecture 3	Basic Data Structures(K maximum sum from two arrays)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	String declaration and manipulation, K maximum sum from two arrays, Missing characters to make a string Pangram, Rearrange characters so that no two adjacent characters are same, Remove minimum number of characters so that two strings become anagram	The students will be able understand in depth about the fundamentals of strings and will be able to solve problems based on the same.	Live demonstration on compiler	
		Basic Data Structures (Missing characters to make a string Pangram, Rearrange characters so that no two adjacent characters are same)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	String declaration and manipulation, K maximum sum from two arrays, Missing characters to make a string Pangram, Rearrange characters so that no two adjacent characters are same, Remove minimum number of characters so that two strings become anagram	The students will be able understand in depth about the fundamentals of strings and will be able to solve problems based on the same.	Live demonstration on compiler	
		Basic Data Structures (Remove minimum number of characters so that two strings become anagram)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	String declaration and manipulation, K maximum sum from two arrays, Missing characters to make a string Pangram, Rearrange characters so that no two adjacent characters are same, Remove minimum number of characters so that two strings become anagram	The students will be able understand in depth about the fundamentals of strings and will be able to solve problems based on the same.	Live demonstration on compiler	

Week 2	Lecture 4	Basic Data Structures (Creation of stack using arrays, Creation of arrays using linked list)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Creation of stack using arrays, Creation of arrays using linked list, Creation of queue using array, Creation of queue using linked list, Implement two Stacks in an array, Implement Stack using Queues, Design a stack with operations on middle element, Implementation of Deque using circular array, Circular Queue, Reversing a Queue	The student will be able to have a good hold on the basic data structures such as stacks, queues as well.	Live demonstration using a compiler	
		Basic Data Structures (Creation of queue using array, Creation of queue using linked list)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Creation of stack using arrays, Creation of arrays using linked list, Creation of queue using array, Creation of queue using linked list, Implement two Stacks in an array, Implement Stack using Queues, Design a stack with operations on middle element, Implementation of Deque using circular array, Circular Queue, Reversing a Queue	The student will be able to have a good hold on the basic data structures such as stacks, queues as well.	Live demonstration using a compiler	

Week 2	Lecture 4	Basic Data Structures (Implement two Stacks in an array)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Creation of stack using arrays, Creation of arrays using linked list, Creation of queue using array, Creation of queue using linked list, Implement two Stacks in an array, Implement Stack using Queues, Design a stack with operations on middle element, Implementation of Deque using circular array, Circular Queue, Reversing a Queue	The student will be able to have a good hold on the basic data structures such as stacks, queues as well.	Live demonstration using a compiler	
		Basic Data Structures (Implement Stack using Queues)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Creation of stack using arrays, Creation of arrays using linked list, Creation of queue using array, Creation of queue using linked list, Implement two Stacks in an array, Implement Stack using Queues, Design a stack with operations on middle element, Implementation of Deque using circular array, Circular Queue, Reversing a Queue	The student will be able to have a good hold on the basic data structures such as stacks, queues as well.	Live demonstration using a compiler	

Week 2	Lecture 4	Basic Data Structures (Design a stack with operations on middle element)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Creation of stack using arrays, Creation of arrays using linked list, Creation of queue using array, Creation of queue using linked list, Implement two Stacks in an array, Implement Stack using Queues, Design a stack with operations on middle element, Implementation of Deque using circular array, Circular Queue, Reversing a Queue	The student will be able to have a good hold on the basic data structures such as stacks, queues as well.	Live demonstration using a compiler	
		Basic Data Structures (Implementation of Deque using circular array, Circular Queue)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Creation of stack using arrays, Creation of arrays using linked list, Creation of queue using array, Creation of queue using linked list, Implement two Stacks in an array, Implement Stack using Queues, Design a stack with operations on middle element, Implementation of Deque using circular array, Circular Queue, Reversing a Queue	The student will be able to have a good hold on the basic data structures such as stacks, queues as well.	Live demonstration using a compiler	

Week 2	Lecture 4	Basic Data Structures (Reversing a Queue)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Creation of stack using arrays, Creation of arrays using linked list, Creation of queue using array, Creation of queue using linked list, Implement two Stacks in an array, Implement Stack using Queues, Design a stack with operations on middle element, Implementation of Deque using circular array, Circular Queue, Reversing a Queue	The student will be able to have a good hold on the basic data structures such as stacks, queues as well.	Live demonstration using a compiler	
Week 3	Lecture 5	Basic Data Structures(Two pointer technique depicted by problem to find if there exists a pair in an unsorted array whose sum is equal to X)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	Two pointer technique depicted by problem to find if there exists a pair in an unsorted array whose sum is equal to X, Count subarrays having an equal sum of elements at even and odd positions, Finding a peak element in array, Find subarray of length K with maximum Peak, Replace every element of the array with the previous element, Rearrange positive and negative numbers	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	

Week 3	Lecture 5	Basic Data Structures(Count subarrays having an equal sum of elements at even and odd positions)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	Two pointer technique depicted by problem to find if there exists a pair in an unsorted array whose sum is equal to X, Count subarrays having an equal sum of elements at even and odd positions, Finding a peak element in array, Find subarray of length K with maximum Peak, Replace every element of the array with the previous element, Rearrange positive and negative numbers	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	
		Basic Data Structures (Finding a peak element in array)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	Two pointer technique depicted by problem to find if there exists a pair in an unsorted array whose sum is equal to X, Count subarrays having an equal sum of elements at even and odd positions, Finding a peak element in array, Find subarray of length K with maximum Peak, Replace every element of the array with the previous element, Rearrange positive and negative numbers	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	

Week 3	Lecture 5	Basic Data Structures(Find subarray of length K with maximum Peak)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	Two pointer technique depicted by problem to find if there exists a pair in an unsorted array whose sum is equal to X, Count subarrays having an equal sum of elements at even and odd positions, Finding a peak element in array, Find subarray of length K with maximum Peak, Replace every element of the array with the previous element, Rearrange positive and negative numbers	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	
		Basic Data Structures (Replace every element of the array with the previous element)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	Two pointer technique depicted by problem to find if there exists a pair in an unsorted array whose sum is equal to X, Count subarrays having an equal sum of elements at even and odd positions, Finding a peak element in array, Find subarray of length K with maximum Peak, Replace every element of the array with the previous element, Rearrange positive and negative numbers	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	

Week 3	Lecture 5	Basic Data Structures (Rearrange positive and negative numbers)	R-1 R-2	RW-1 RW-2 RW-3 RW-5	Two pointer technique depicted by problem to find if there exists a pair in an unsorted array whose sum is equal to X, Count subarrays having an equal sum of elements at even and odd positions, Finding a peak element in array, Find subarray of length K with maximum Peak, Replace every element of the array with the previous element, Rearrange positive and negative numbers	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	
	Lecture 6	Basic Data Structures (Rearrange array such that even index elements are smaller and odd index elements are greater)	R-1 R-2	RW-1 RW-3 RW-4 RW-5	Rearrange array such that even index elements are smaller and odd index elements are greater, Find the first non-repeating element in a given array of integers, Find the majority element, Count strings with consecutive ones, Check if all bits can be made same by single flip, Last Moment Before All Ants Fall Out of a Plank, Move All Zeros to the End of Array	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	

Week 3	Lecture 6	Basic Data Structures(Find the first non-repeating element in a given array of integers)	R-1 R-2	RW-1 RW-3 RW-4 RW-5	Rearrange array such that even index elements are smaller and odd index elements are greater, Find the first non-repeating element in a given array of integers, Find the majority element, Count strings with consecutive ones, Check if all bits can be made same by single flip, Last Moment Before All Ants Fall Out of a Plank, Move All Zeros to the End of Array	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	
		Basic Data Structures(Find the majority element)	R-1 R-2	RW-1 RW-3 RW-4 RW-5	Rearrange array such that even index elements are smaller and odd index elements are greater, Find the first non-repeating element in a given array of integers, Find the majority element, Count strings with consecutive ones, Check if all bits can be made same by single flip, Last Moment Before All Ants Fall Out of a Plank, Move All Zeros to the End of Array	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	

Week 3	Lecture 6	Basic Data Structures(Count strings with consecutive ones)	R-1 R-2	RW-1 RW-3 RW-4 RW-5	Rearrange array such that even index elements are smaller and odd index elements are greater, Find the first non-repeating element in a given array of integers, Find the majority element, Count strings with consecutive ones, Check if all bits can be made same by single flip, Last Moment Before All Ants Fall Out of a Plank, Move All Zeros to the End of Array	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	
		Basic Data Structures(Check if all bits can be made same by single flip)	R-1 R-2	RW-1 RW-3 RW-4 RW-5	Rearrange array such that even index elements are smaller and odd index elements are greater, Find the first non-repeating element in a given array of integers, Find the majority element, Count strings with consecutive ones, Check if all bits can be made same by single flip, Last Moment Before All Ants Fall Out of a Plank, Move All Zeros to the End of Array	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	

Week 3	Lecture 6	Basic Data Structures(Last Moment Before All Ants Fall Out of a Plank, Move All Zeros to End of Array)	R-1 R-2	RW-1 RW-3 RW-4 RW-5	Rearrange array such that even index elements are smaller and odd index elements are greater, Find the first non-repeating element in a given array of integers, Find the majority element, Count strings with consecutive ones, Check if all bits can be made same by single flip, Last Moment Before All Ants Fall Out of a Plank, Move All Zeros to the End of Array	Students will be able to solve complex problems based upon array.	Live demonstration using a compiler	
Week 4	Lecture 7	Basic math operations (addition, subtraction, multiplication, division and exponentiation)(Fast modulo multiplication)	R-1 R-2	RW-5	Fast modulo multiplication, Exponential squaring, N-th non-square number	The student will have a hands on practice and a good hold on the basic math operations	Live demonstration on compiler	
		Basic math operations (addition, subtraction, multiplication, division and exponentiation)(Exponential squaring)	R-1 R-2	RW-5	Fast modulo multiplication, Exponential squaring, N-th non-square number	The student will have a hands on practice and a good hold on the basic math operations	Live demonstration on compiler	
		Basic math operations (addition, subtraction, multiplication, division and exponentiation)(N-th non-square number)	R-1 R-2	RW-5	Fast modulo multiplication, Exponential squaring, N-th non-square number	The student will have a hands on practice and a good hold on the basic math operations	Live demonstration on compiler	
	Lecture 8	Basic math operations (addition, subtraction, multiplication, division and exponentiation)(Modular Exponentiation)	R-1 R-2	RW-5	Modular Exponentiation, Modular multiplicative inverse, Euler's Totient Function	The student will have a hands on practice and a good hold on the basic math operations	Live demonstration on compiler	
		Basic math operations (addition, subtraction, multiplication, division and exponentiation)(Modular multiplicative inverse)	R-1 R-2	RW-5	Modular Exponentiation, Modular multiplicative inverse, Euler's Totient Function	The student will have a hands on practice and a good hold on the basic math operations	Live demonstration on compiler	

Week 5	Lecture 9	Basic math operations (addition, subtraction, multiplication, division and exponentiation)(Sum of middle row and element in matrix)	R-1 R-2	RW-5	Mobius and divisor functions ,Sum of middle row and element in matrix, Checking if all rows of a matrix are circular rotations of each other	The student will have a hands on practice and a good hold on the basic math operations	Live demonstration on compiler	
		Basic math operations (addition, subtraction, multiplication, division and exponentiation)(Checking if all rows of a matrix are circular rotations of each other)	R-1 R-2	RW-5	Mobius and divisor functions ,Sum of middle row and element in matrix, Checking if all rows of a matrix are circular rotations of each other	The student will have a hands on practice and a good hold on the basic math operations	Live demonstration on compiler	
	Lecture 10	GCD and Primality testing (Basic Euclidean algorithm)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Basic Euclidean algorithm, Extended Euclidean algorithm, Total number of divisors of a number, Finding all prime factors of a number	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
		GCD and Primality testing (Extended Euclidean algorithm)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Basic Euclidean algorithm, Extended Euclidean algorithm, Total number of divisors of a number, Finding all prime factors of a number	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
		GCD and Primality testing (Total number of divisors of a number)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Basic Euclidean algorithm, Extended Euclidean algorithm, Total number of divisors of a number, Finding all prime factors of a number	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
		GCD and Primality testing (Finding all prime factors of a number)	R-1 R-2	RW-1 RW-2 RW-3 RW-4 RW-5	Basic Euclidean algorithm, Extended Euclidean algorithm, Total number of divisors of a number, Finding all prime factors of a number	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
Week 6	Lecture 11				Test- Situation based problem solving 1			

Week 6	Lecture 12	GCD and Primality testing (K-jagged numbers)	R-1 R-2	RW-5	K-jagged numbers, Stormer numbers, Finding the prime factors by taking the square root, P-smooth numbers in given ranges	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
		GCD and Primality testing (Stormer numbers)	R-1 R-2	RW-5	K-jagged numbers, Stormer numbers, Finding the prime factors by taking the square root, P-smooth numbers in given ranges	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
		GCD and Primality testing (Finding the prime factors by taking the square root)	R-1 R-2	RW-5	K-jagged numbers, Stormer numbers, Finding the prime factors by taking the square root, P-smooth numbers in given ranges	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
		GCD and Primality testing (P-smooth numbers in given ranges)	R-1 R-2	RW-5	K-jagged numbers, Stormer numbers, Finding the prime factors by taking the square root, P-smooth numbers in given ranges	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
Week 7	Lecture 13	GCD and Primality testing (Frugal number)	R-1 R-2	RW-2 RW-4	P-smooth numbers in given ranges, Problems based on GCD and primality testing	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
		GCD and Primality testing (Lemoine's Conjecture)	R-1 R-2	RW-2 RW-4	P-smooth numbers in given ranges, Problems based on GCD and primality testing	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
		GCD and Primality testing (Problems based on GCD and primality testing)	R-1 R-2	RW-2 RW-4	P-smooth numbers in given ranges, Problems based on GCD and primality testing	Students will get hands on practice and a good hold on GCD as well as primality testing	Live demonstration on compiler	
		SPILL OVER						
Week 14	Lecture 28				Spill Over			
		MID-TERM						

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0202 (2 out of 2)

Component	Weightage (%)	Mapped CO(s)
Recoup CA		
Test- Situation based problem solving 1	50	
Test- Situation based problem solving 2	50	

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Recoup CA	subjective test based on the syllabus	Additional chance of CA	Individual	Online	30	8 / 10
Test- Situation based problem solving 1	To build the problem solving skills of the students and to evaluate competency in the same	The student will be given some situation based problems for which he/she is expected to write code for optimally solving the same. The academic task is aimed at checking the knowledge gained till week 5.	Individual	Online	30	5 / 6
Test- Situation based problem solving 2	To build the problem solving skills of the students and to evaluate competency in the same	The student will be given some situation based problems for which he/she is expected to write code for optimally solving the same. The academic task is aimed at checking the knowledge gained till week 11.	Individual	Online	30	11 / 12

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	Practicals	Program to insert an element in array		Student will be given hands on practice on array insertion and deletion
	Practicals	Program to delete an element from an array		Student will be given hands on practice on array insertion and deletion
Practical 2	Practicals	Program to perform array arrangement		The students will be having hands on practice on array rearrangement
Practical 3	Practicals	Program to print a matrix in spiral form		The students will be given hand on practice on matrices
Practical 4	Practicals	Program to find K maximum sum from two arrays		The students will be given hands on practice on array problems
Practical 5	Practicals	Program to find missing characters to make a string Pangram		The students will be having hands on practice on problems based on character arrays and string
Practical 6	Practicals	Program to create a stack using arrays		The students will get hands on practice on linear data structures using arrays, stacks and queues
Practical 7	Practicals	Program to print the reverse of a queue		The students will get hands on practice on linear data structures using arrays, stacks and queues
Practical 8	Practicals	Program to print peak element in array		The students will get hands on practice on linear data structures namely arrays
Practical 9	Practicals	Program to print positive and negative numbers		The students will get hands on practice on linear data structures namely arrays
Practical 10	Practicals	Program to print total divisors of a number		The students will get hands on practice on linear data structures namely arrays
Practical 11	Practicals	Program to print K jagged numbers		The students will get hands on practice on finding divisors of a number, printing K jagged numbers etc.
Practical 12	Practicals	Program for finding minimum product subset of an array		The students will get hands on practice on problems belonging to the greedy algorithmic paradigm
Practical 13	Practicals	Program to perform activity selection problem		The students will get hands on practice on problems belonging to the greedy algorithmic paradigm
Practical 14	Practicals	Program to find a pair with a given difference		The students will get hands on practice on problems belonging to the greedy algorithmic paradigm
	SPILL OVER			
Practical 15	Spill Over			

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